

Mathematics Extended Essay on the topic of Erdős' Probabilistic method

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1 Introduction

- A brief introduction - My motivation to explore this topic

To briefly help the reader to understand this essay better, below are a few definitions which might come in handy throughout: An Object is a

2 Scale Free Networks and Graph Theory

- Describing the concept of graphs - Defining terms - Applying the idea to scale free networks

Mathematics is more than just numbers and algebra. One predominant branch which is often ignored is called combinatorics. It focuses on the idea of counting, which is applied in order to find a result but can also define a final answer. I will talk more about that concept using an example below. Part of combinatorics is the concept of graphs and its related graph theory which explores the relationship between objects. These should not be confused with a graph on a cartesian plane. These graphs allow the mathematical analysis of networks and connections, like streets, mobile networks, social media networks or From now on the use of the word graph will refer to the mathematical graph which makes part of discrete mathematics, i.e. a network of nodes connected by edges.

3 The Probabilistic Method

- Describing Probabilistic Notation - Applying using simple a simple Ramsey number example ($R(3,3) = 6$)

As we discussed above, graph theory is computationally difficult to explore. When trying to generate every possible combination of worlds

4 Predicting clique sizes in Scale Free Networks

- Setting out the problem: we want to find specific clique sizes in a scale free network with 4 hubs - Using the Chung-Lu Random Graph Model, we will describe how to statistically generate a graph, with different probabilities of nodes and edges occurring. - Using the first moment we will explain the concept of

5 Building Scale Free Networks

- Using the help of AI to code a python script, I aim to generate a statistically meaningful number of Scale-Free Networks. - By analysing the sizes of clique's present, I should be able to find similar results to my theoretical/probabilistic approach

6 Comparing Results

- Comparing results and making sure theory aligns with empirical data

7 Conclusion

- Conclude on the uses of my results